End of Year Maths Assessment – Year 6

Number and Place Value

1. Write the following numbers in words.

6,540,100 _____

8,111,093 _____

2. Write the follow numbers in digits.

Three hundred and sixty-five thousand, nine hundred and two. _____

Four million, eight hundred and ten thousand, four hundred and seventy-two. _____

3. Use the symbols > or < to compare these numbers.

5,645,011 ☐ **5,445,011** **1,087,998** ☐ **1,087,990** **987,890** ☐ **987,990**

4. Order the following numbers from smallest to largest.

909,990	990,009	900,999	909,999	900,009

5. Round the following numbers to the nearest 10.

14 _____ **35** _____ **68** _____

Round the following numbers to the nearest 100.

211 _____ **765** _____ **985** _____

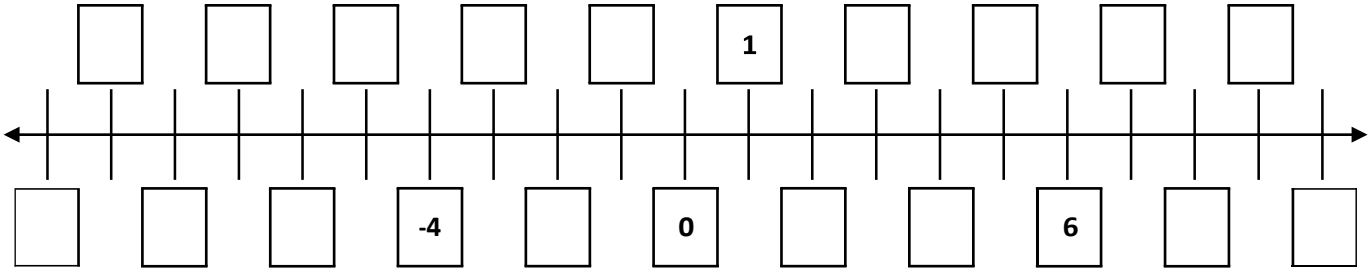
Round the following numbers to the nearest 1,000.

3,987 _____ **2,502** _____ **3,456** _____

Round the following numbers to the nearest 10,000.

15,987 _____ **23,450** _____ **32,099** _____

6. Fill in the missing numbers on the number line.



7. Put these temperatures in order of coolest to warmest.

- 5°C	0°C	- 7°C	5°C	- 15°C

8. Work out the mystery number. The mystery number has 6 digits.

- **The thousands digit is 8.**
- **The hundreds digit is half of the thousands digit.**
- **The ones digit is a quarter of the thousands digit.**
- **The hundred thousands digit is the same as the ones digit.**
- **The tens digit is 3^2.**
- **The ten thousands digit is a prime number bigger than 5 and smaller than 10.**

The mystery number is: ____ ____ ____ ____ ____ ____

Addition, Subtraction, Multiplication and Division

9. 3,467 x 24 =

10. 5,324 x 32 =

11. 4,608 ÷ 18 =

12. 7,056 ÷ 25 =

13. 7 x ☐ = 1,400 ☐ x 300 = 9,000

350 ÷ ☐ = 7 ☐ ÷ 6 = 600

14. Find the common factors of 12 and 30.

15. Find the common multiples of 9 and 12 under 100.

16. List the first 5 prime numbers after 10.

☐ ☐ ☐ ☐ ☐

17. Complete the calculations.

36 + (8 x 9) = _____ 6^2 + 3 + 9 x 2 = _____

(36 + 8) x 9 = _____ 6^2 + (3 + 9) x 2 = _____

18. John weighs 57kg. Alice weighs 16kg less than John. Haroon weighs 5kg more than Alice.

What is the combined weight of John, Alice and Haroon?

19. Hallie has £10 to spend in the shop. She buys 2 pencils which are 75p each, a calculator which is £2.50, she buys 4 books which are £1.10 each and a pack of pens for £0.99. How much change does she get?

20. A plane travels 8,746 miles on Monday, 15,454 miles on Tuesday, 4,587 miles on Wednesday, 7,123 miles on Thursday and 896 miles on Friday. Estimate how many miles the plane flew over the 5 days.

Fractions (including decimals and percentages)

21. Simplify the fractions into the simplest form.

$\frac{12}{36} = ___$ $\frac{10}{90} = ___$ $\frac{3}{12} = ___$ $\frac{9}{45} = ___$ $\frac{14}{56} = ___$

22. Order these fractions from smallest to largest.

$\frac{7}{8}$ $\frac{2}{3}$ $\frac{3}{4}$ $\frac{2}{5}$ $\frac{1}{2}$

☐ ☐ ☐ ☐ ☐

23. Compare these fractions using the correct symbol. <, > or =

$\frac{2}{3}$ ☐ $\frac{7}{12}$ $\frac{2}{5}$ ☐ $\frac{4}{10}$ $\frac{2}{3}$ ☐ $\frac{7}{9}$

24. Complete the following calculations.

$\frac{1}{2} + \frac{3}{8} =$ ☐ $\frac{6}{21} + \frac{3}{7} =$ ☐

$1\frac{2}{5} - \frac{11}{15} =$ ☐ $1\frac{3}{7} - \frac{1}{2} =$ ☐

25. Multiply the following fractions. Write the answers in the simplest form.

$\frac{2}{3} \times \frac{1}{3} =$ ☐ $\frac{1}{4} \times \frac{2}{5} =$ ☐ $\frac{3}{7} \times \frac{1}{3} =$ ☐

26. Divide the following fractions.

$\frac{3}{4} \div 4 = \square$ $\frac{2}{5} \div 3 = \square$ $\frac{3}{8} \div 5 = \square$

27. Draw lines to match the fraction to the equivalent decimal.

0.1	$\frac{3}{10}$
0.3	$\frac{1}{4}$
0.25	$\frac{4}{5}$
0.6	$\frac{6}{10}$
0.8	$\frac{1}{10}$

28. Complete the table below.

	x 10	x 100	x 1,000
1.554			
0.8			
	÷ 10	÷ 100	÷ 1,000
65			
137			

29. Circle the incorrect calculation.

1.36 x 4 = 5.44 2.32 x 6 = 13.85 1.74 x 5 = 8.7

30. 34.8 ÷ 2 =

31. 597.3 ÷ 3 =

32. Carla has collected £553 in sponsorship doing a charity bike ride. She wants to share the money equally between 4 different charities.

How much will each charity get?

33. Emmie runs 7 laps of the football pitch which is 950 metres. How many metres is 1 lap to the nearest metre?

34. Charlie shares his cake with his friends. He gives 25% to William, 0.2 of the cake to Suzie and $\frac{1}{10}$ of the cake to Monty. Charlie gets the rest of the cake. What percentage does Charlie get?

Ratio and Proportion

Use the information below to answer questions 35 and 36.

Greengrass Greengrocers
Carrots - £1.40 per kg
Apples – 4 for £1
Potatoes - £2.25 for 1.5kg bag
Cheese – 60p per 100g

35. Johan buys 2kg of carrots, 2 apples and 500g of potatoes. How much does he pay altogether?

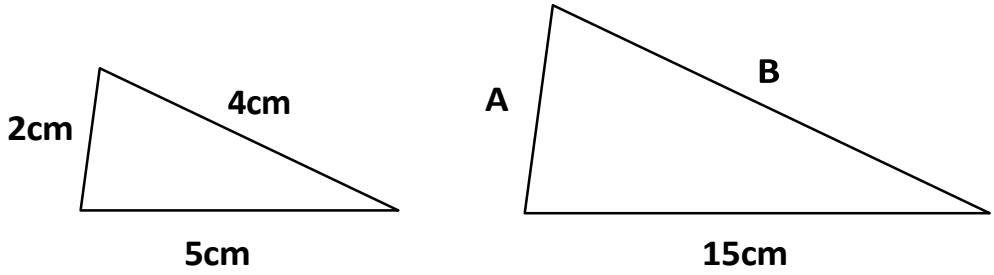
36. Alicia needs 250g of cheese. How much will it cost?

37. Calculate the following percentages.

15% of 200 =	_____	**5% of 80 =**	_____
30% of 75 =	_____	**60% of 350 =**	_____

38. Joe buys a new pair of trainers for £52. He has a 15% off voucher. What is the price of the trainers after he uses the voucher?

39. These shapes are similar. Find the missing lengths.



A = _____ B = _____

What is the scale factor?_____

40. Taku and Arianna have a car-boot sale. They make £240 in total. They share the money between them both but Taku gets £20 more than Arianna. How much do they both get?

Taku gets_____and Arianna gets _____.

Algebra

41. Work out the value of each of the letters.

$a + 17 = 42$	$2n - 16 = 32$	$5d = 50$
$a =$ _____	$n =$ _____	$d =$ _____

42. Create a sequence by completing the table.

n	1	2	3	4	5	6	7	8	9
2n	2	4							
3n + 1	4	7							

43. Work out the missing numbers in each sequence and write the rule.

5 11 [ ] 23 29 [ ] 41 [ ] 53 [ ]

a. The rule is _____

[ ] 50 35 [ ] 5 [ ] -25 [ ] -55 [ ]

b. The rule is _____

2 [ ] 5 9 17 [ ] 65 [ ]

c. The rule is _____

44. Tim is thinking of a number. He multiplies the number by 3 and then adds 21. He gets an answer of 45. What is the number he is thinking of?

45. Ashleigh is thinking of a number. She divides the number by 4 and then adds 11. She gets an answer of 22. What is the number she is thinking of?

46. Find three different solutions to this equation. Numbers must be whole numbers.

$$3a + 2b = 24$$

a = and b =	a = and b =	a = and b =

47. Two numbers when added together make 37. Both numbers are less than 20. What are the two numbers?

and

Measurement

48. Shania is filling up a paddling pool. The pool holds 7.2 litres of water. She is using a cup which holds 120ml.

How many cups full of water will it take to fill the paddling pool?

49. Convert the following litres to millilitres and millilitres to litres.

$1\frac{1}{2}$ l = _____ml $3\frac{1}{4}$ l = _____ ml

1,350 ml = _____l 1,765 ml = _____l

50. Convert the following kilograms to grams and grams to kilograms.

$2\frac{3}{4}$ kg = _____g $5\frac{1}{2}$ kg = _____ g

12,420 g = _____kg 7,650 g = _____kg

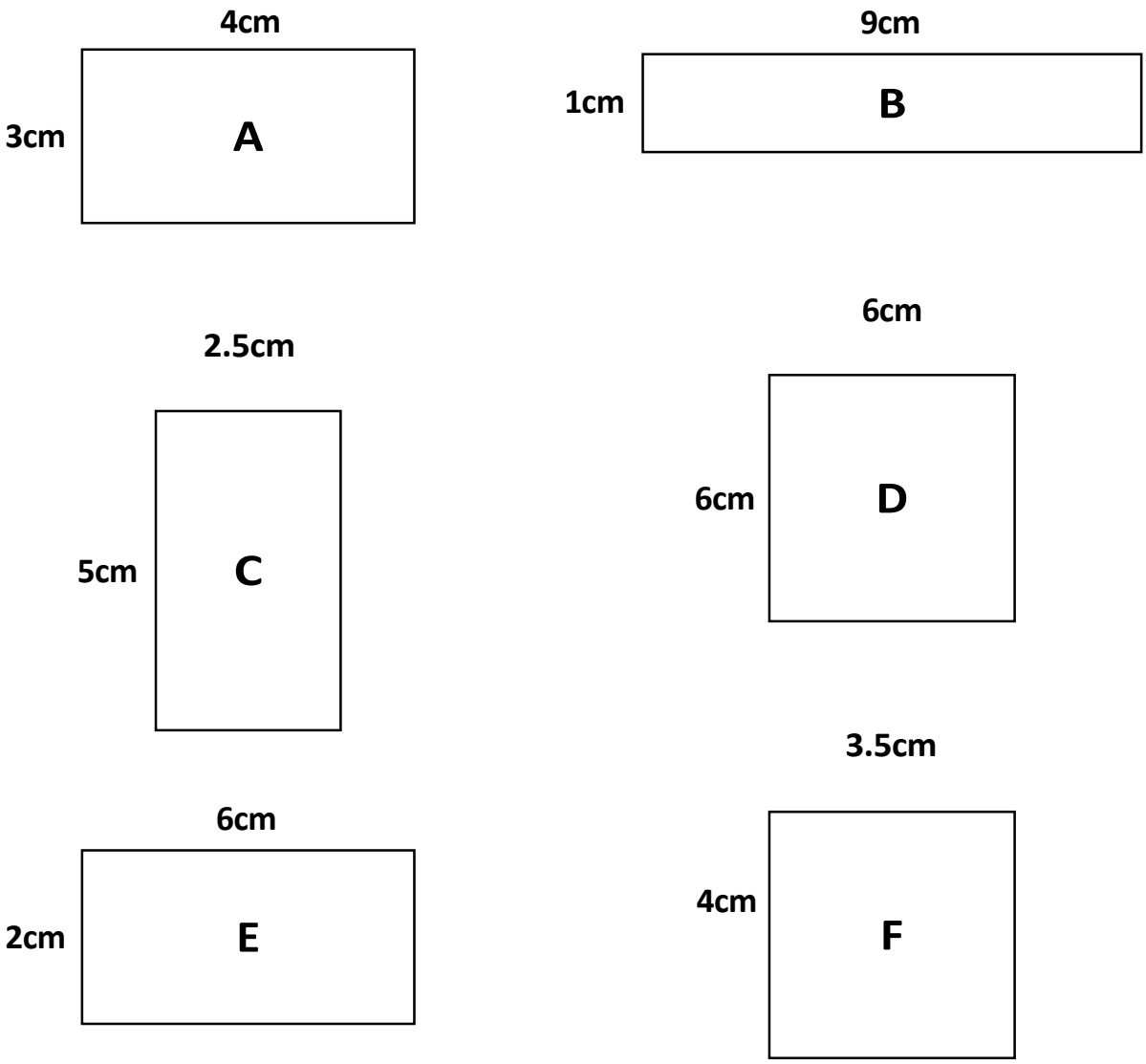
51. Complete the table.

mm	cm	m
	50cm	
		3.5m
250mm		
	310cm	

52. Complete the table.

Approximately 1 mile = 1.6 kilometres	
3 miles	km
miles	8 km
6 miles	km
miles	0.8 km

53. Look at the shapes below.



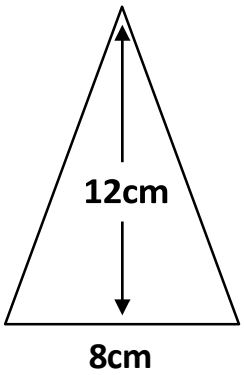
Which two shapes have the same area?

Which two shapes have the same perimeter?

54. The area of a rectangle is 102cm². One side is 6cm, what is the length of the other side? _____

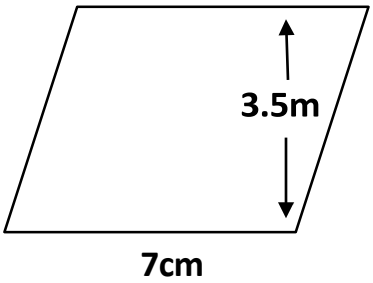
55. The area of a square is 81m². What is the length of one side? _____

56. Calculate the area of the triangle.



Area =

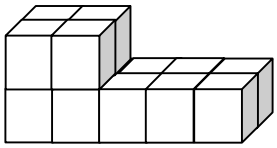
57. Calculate the are of the parallelogram.



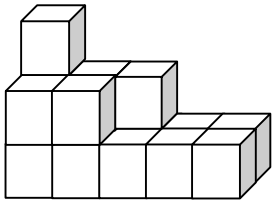
Area =

58. Calculate the volume of the three shapes. Each cube is 1cm³.

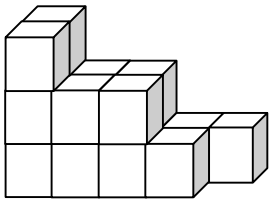
A



B



C



Volume = _____ **Volume = _____** **Volume = _____**

Which shape has the largest volume? _____

Geometry (properties of shapes)

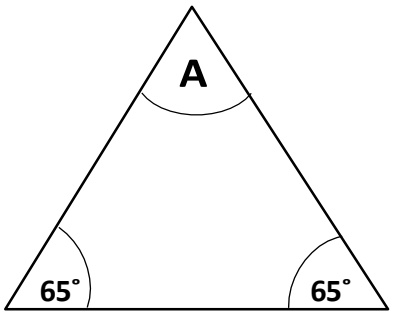
59. Draw a three sided 2D shape which has 2 acute angles, 1 right-angle and 2 sides are equal.

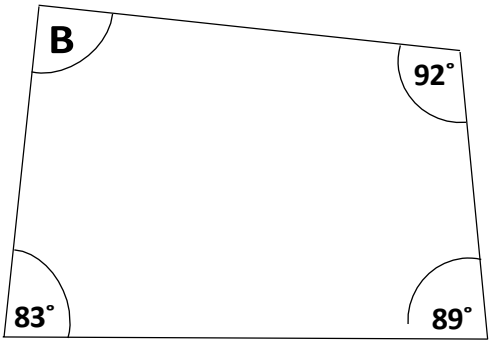
60. Draw a five sided shape which has 2 right-angles.

61. Fill in the properties for the 3D shapes.

3D shape	Faces	Vertices	Edges
Cylinder			
Sphere			
Tetrahedron			
Square Based pyramid			

62. Find the missing angles.

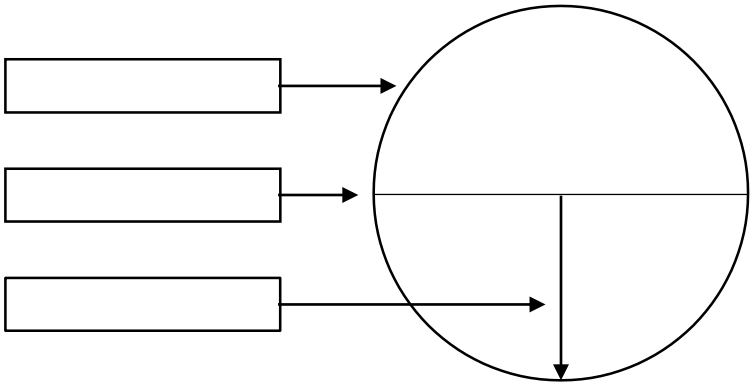




A = _____

B = _____

63. Label the parts of the circle.

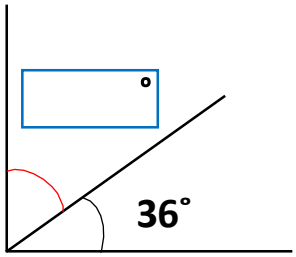


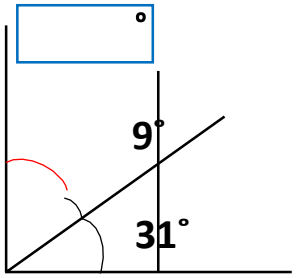
Circumference

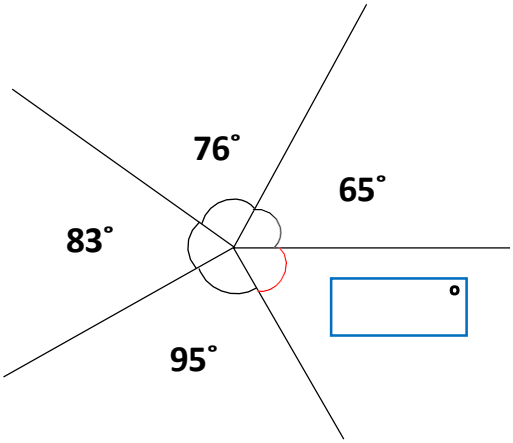
Radius

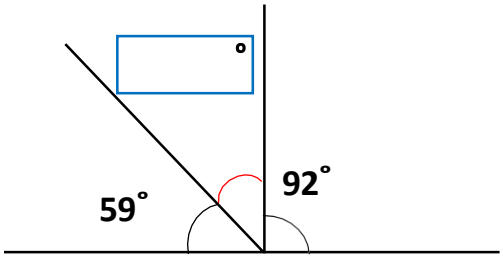
Diameter

64. Calculate the missing angles.



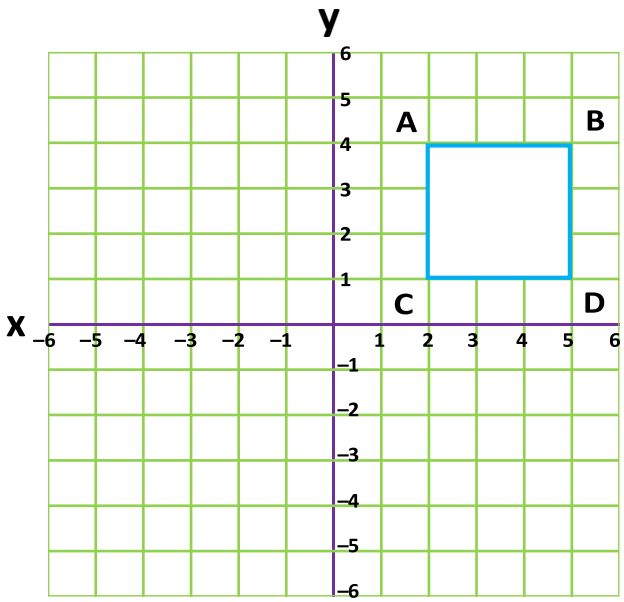






Geometry (position and direction)

65. Write the co-ordinates of the square.



A = _____

B = _____

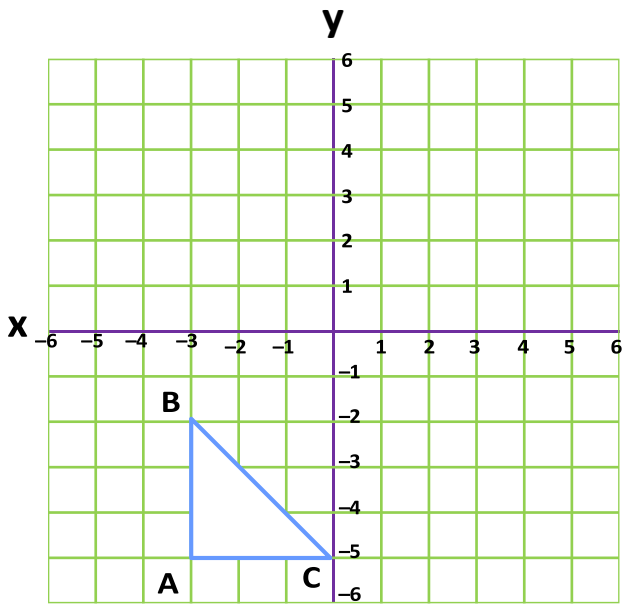
C = _____

D = _____

66. Reflect the square on the y axis and write the new co-ordinates.

A = _____ B = _____ C = _____ D_____

67. Translate the shape.



Instructions

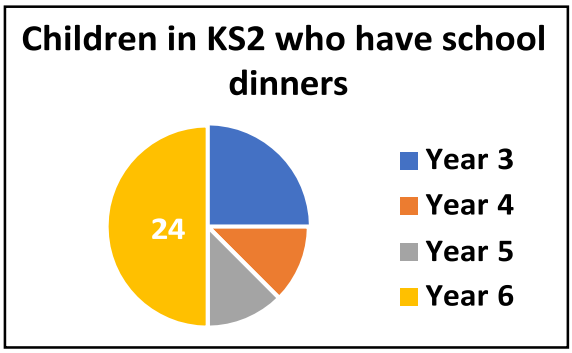
- **Translate the shape 2 squares to the right and 6 squares up.**

68. What are the new co-ordinates

A = _____ B = _____ C = _____

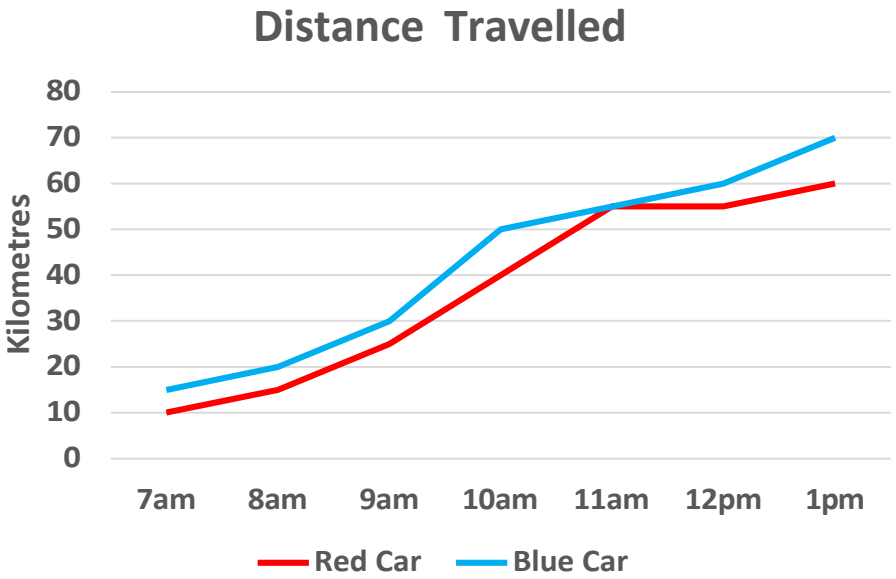
Statistics

69. Use the information from the pie chart.



How many children have school dinners in Year 3? _____
How many children have school dinners in Year 5? _____
How many children in total have school dinners? _____
What fraction of children asked have school dinners in year 4? _____

70.



How far did the red car travel from 7am to 9am? _____

Which car travelled the furthest between 12pm and 1pm?_____

How far did the blue car travel from 10 am to 1pm? _____

How long did it take the blue car to travel 60km from 20km? _____

71. Children in Year 6 are gathering the ages of their parents.

27	37
31	29
31	32
30	29
34	30

What is the mean age? _____